

Last-Mile Delivery Performance Dataset

Project: Delivery Analytics & Delay Root Cause Analysis
Dataset Size: 5,200 Records | 23 Columns | Jan 2024 – Dec 2024

Dataset Summary

Metric	Value	Metric	Value
Total Orders	5,200	Total Cities	15
Delayed Orders	2,236 (43.0%)	Delivery Agents	150
SLA Breached	1,459 (28.1%)	Avg Delay (when delayed)	3.9 days
Date Range	Jan – Dec 2024	Vendors	8
Total Order Value	₹3.88 Cr	Product Categories	8

Column Description

Column Name	Data Type	Description	Example Value
Order_ID	String	Unique order identifier	ORD100000
Order_Date	DateTime	Date and time when order was placed	2024-03-15 14:30
Promised_Delivery_Date	Date	Date promised to customer at checkout	2024-03-17
Actual_Delivery_Date	Date	Date order was actually delivered	2024-03-19
City	String	Delivery destination city (15 Indian cities)	Mumbai, Delhi, Bangalore...
Zone	String	Geographic zone (North/South/East/West/Central)	South
Warehouse_ID	String	Source warehouse for the order	WH_BLR_01
Delivery_Agent_ID	String	Unique agent assigned to deliver (150 agents)	AGT_0042
Vendor	String	Logistics vendor/partner used	Delhivery, BlueDart, Ekart...
Product_Category	String	Category of the product ordered	Electronics, Fashion...
Order_Value_INR	Float	Total order value in Indian Rupees	2450.75
Payment_Mode	String	Payment method - Prepaid or COD	COD
Promised_Delivery_Days	Integer	Number of days promised for delivery	2

Actual_Delivery_Days	Integer	Actual number of days taken for delivery	4
Delay_Days	Integer	Number of days delayed (0 = on time)	2
Is_Delayed	String (Y/N)	Whether order was delayed beyond promised date	Yes
Delay_Reason	String	Root cause of delay (None if on time)	Traffic Congestion
SLA_Breached	String (Y/N)	SLA breached if delay > 2 days	No
Delivery_Attempts	Integer	Number of delivery attempts made	1
Month	String	Month of order placement	October
Quarter	String	Quarter of order placement	Q4
Is_Festive_Season	String (Y/N)	Oct-Nov flagged as festive season	Yes
Is_Monsoon_Season	String (Y/N)	Jun-Sep flagged as monsoon season	No

SQL Analysis Questions You Can Solve

#	Business Question	SQL Concepts Used
1	Which cities have the highest delay rate? Rank them.	GROUP BY, COUNT, RANK()
2	Which warehouse has maximum SLA breaches per month?	Window Functions, CTE
3	Top 10 delivery agents by delay rate (min 50 orders)	HAVING, DENSE_RANK()
4	Compare delay rate: Festive vs Non-Festive season	CASE WHEN, GROUP BY
5	Which vendor has best on-time delivery % by zone?	Subquery, PARTITION BY
6	Month-over-month trend in SLA breaches	LAG(), Window Functions
7	Does COD have higher delay rate than Prepaid?	Conditional Aggregation
8	Product categories with highest avg delay days	AVG(), ORDER BY
9	Agents who exceeded avg delay in their city (outliers)	CTE + Self Join
10	Root cause analysis: which reason causes most revenue loss?	JOIN, SUM, GROUP BY

Power BI Dashboard Structure

Page	Dashboard Name	Key Visuals	KPIs
1	Executive Overview	KPI Cards, Delay Trend Line, City Map	Total Orders, Delay %, SLA Breach %
2	City & Zone Analysis	Filled Map, Bar Chart by Zone, Heat Map	Delay rate by city, Zone comparison
3	Warehouse Performance	Bar Chart, Matrix Table, Scatter Plot	Orders per WH, Breach count, Efficiency score
4	Agent Performance	Top/Bottom 10 Table, Gauge Chart	On-time %, Avg delay, Attempts per delivery
5	Root Cause Analysis	Donut Chart, Pareto Chart, Treemap	Delay reason %, Revenue impact by reason
6	Seasonal Trends	Line Chart, Clustered Bar, Waterfall	Festive vs Normal, Monsoon impact, MoM trend

Sample Data (First 30 Rows)

Order ID	Order Date	City	Vendor	Product Category	Order Value INR	Payment Mode	Delay Days	Is Delayed	Delay Reason	SLA Breached
ORD100000	2024-02-27 00:47	Surat	BlueDart	Beauty	10198.99	Prepaid	2	Yes	Wrong Address	No
ORD100001	2024-04-21 07:32	Delhi	Ekart	Electronics	3423.54	COD	2	Yes	Warehouse Processing Delay	No
ORD100002	2024-05-22 04:13	Kolkata	Dunzo	Electronics	5480.84	COD	4	Yes	Wrong Address	Yes
ORD100003	2024-07-12 02:35	Delhi	XpressBees	Electronics	8723.68	Prepaid	0	No	nan	No

Order ID	Order Date	City	Vendor	Product Category	Order Value INR	Payment Mode	Delay Days	Is Delayed	Delay Reason	SLA Breached
ORD100004	2024-04-29 03:24	Delhi	Delhivery	Books	2565.41	COD	0	No	nan	No
ORD100005	2024-05-05 05:29	Jaipur	BlueDart	Home & Kitchen	10316.52	Prepaid	6	Yes	Vehicle Breakdown	Yes
ORD100006	2024-05-17 02:13	Pune	Ekart	Toys	3307.5	COD	0	No	nan	No
ORD100007	2024-05-06 23:35	Bangalore	Dunzo	Groceries	13482.67	COD	5	Yes	Warehouse Processing Delay	Yes
ORD100008	2024-01-25 03:09	Patna	DTDC	Toys	9006.67	COD	0	No	nan	No
ORD100009	2024-01-06 21:46	Bhopal	Dunzo	Groceries	11563.69	COD	2	Yes	nan	No
ORD100010	2024-05-14 16:48	Nagpur	Ekart	Home & Kitchen	13077.2	COD	0	No	nan	No
ORD100011	2024-03-23 17:49	Patna	DTDC	Sports	7405.79	Prepaid	0	No	nan	No
ORD100012	2024-05-03 18:05	Mumbai	Dunzo	Groceries	14673.07	Prepaid	0	No	nan	No
ORD100013	2024-05-15 16:55	Bangalore	Ecom Express	Books	13944.37	Prepaid	0	No	nan	No
ORD100014	2024-09-21 14:07	Ahmedabad	BlueDart	Home & Kitchen	8886.38	Prepaid	3	Yes	Traffic Congestion	Yes
ORD100015	2024-11-19 01:14	Nagpur	Dunzo	Books	1202.23	Prepaid	0	No	nan	No
ORD100016	2024-03-08 23:59	Jaipur	Ecom Express	Fashion	11801.6	COD	0	No	nan	No
ORD100017	2024-07-29 14:55	Pune	XpressBees	Fashion	1050.1	COD	0	No	nan	No
ORD100018	2024-10-01 14:08	Hyderabad	Delhivery	Toys	13136.12	Prepaid	4	Yes	Vehicle Breakdown	Yes
ORD100019	2024-10-03 00:05	Surat	Ecom Express	Groceries	3324.15	COD	4	Yes	Route Optimization Failure	Yes
ORD100020	2024-05-15 14:18	Pune	BlueDart	Electronics	2448.73	COD	0	No	nan	No
ORD100021	2024-01-26 18:30	Mumbai	Delhivery	Groceries	994.66	Prepaid	0	No	nan	No
ORD100022	2024-04-30 12:07	Surat	DTDC	Sports	9347.82	COD	5	Yes	Traffic Congestion	Yes
ORD100023	2024-06-09 07:16	Surat	Dunzo	Electronics	6939.78	Prepaid	6	Yes	Weather Conditions	Yes
ORD100024	2024-02-07 17:13	Delhi	DTDC	Toys	13207.16	COD	3	Yes	Wrong Address	Yes
ORD100025	2024-12-07 17:19	Mumbai	DTDC	Groceries	4077.32	Prepaid	0	No	nan	No
ORD100026	2024-04-17 22:21	Lucknow	Delhivery	Electronics	7655.32	COD	0	No	nan	No
ORD100027	2024-05-21 01:00	Bhopal	BlueDart	Beauty	2549.49	COD	0	No	nan	No
ORD100028	2024-12-19 04:34	Delhi	Shadowfax	Books	6532.33	Prepaid	0	No	nan	No
ORD100029	2024-07-02 06:43	Indore	XpressBees	Groceries	11734.73	COD	0	No	nan	No

Note: This dataset is synthetically generated for portfolio/learning purposes using realistic Indian e-commerce delivery patterns. All Order IDs, Agent IDs are fictional.